

# Customer Segmentation and Prediction

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## Customer Segmentation and Prediction

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# Project Purpose

- Segment customers based on purchasing patterns using RFM (Recency, Frequency, Monetary) analysis.
- Build a predictive model to assign new customers to relevant segments.
- Derive marketing strategies for each segment to increase engagement and sales.

# Dataset Overview

- Data Source: Kaggle
- Size: 4298 customers and 18053 transactions
- Key Variables: Recency, Frequency, Monetary
- Data Cleaning: Removed duplicates & missing values, handled outliers, formatted date & numeric fields

# Methodology



Data Preparation



Feature Engineering (RFM)



Segmentation



Insights & Monetary Analysis



Predictive Modelling

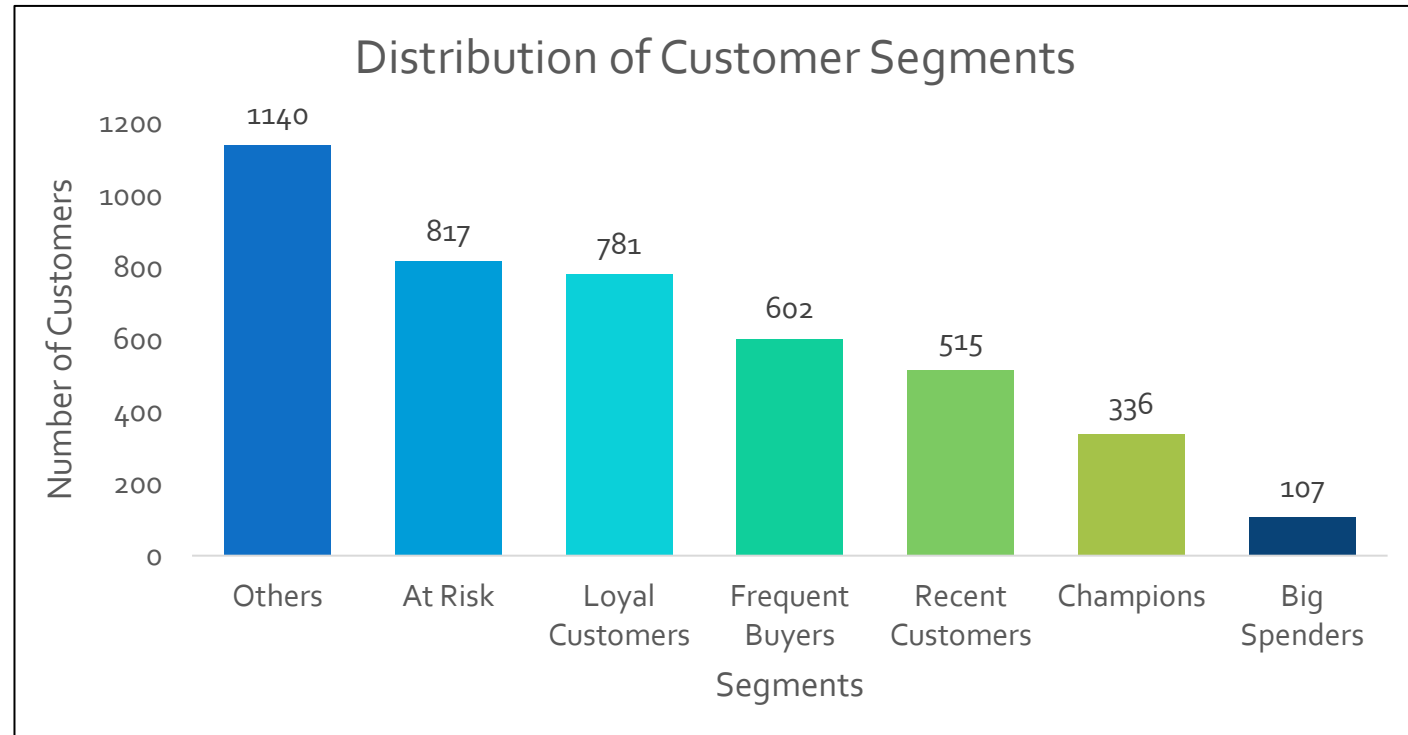


Recommendations & Conclusion

# RFM Segmentation Results

- Total customers analyzed: **4,298**
- Segments identified:
  - **Others:** 1,140 customers (26.5%)
  - **At Risk:** 817 customers (19.0%)
  - **Loyal Customers:** 781 customers (18.2%)
  - **Frequent Buyers:** 602 customers (14.0%)
  - **Recent Customers:** 515 customers (12.0%)
  - **Champions:** 336 customers (7.8%)
  - **Big Spenders:** 107 customers (2.5%)

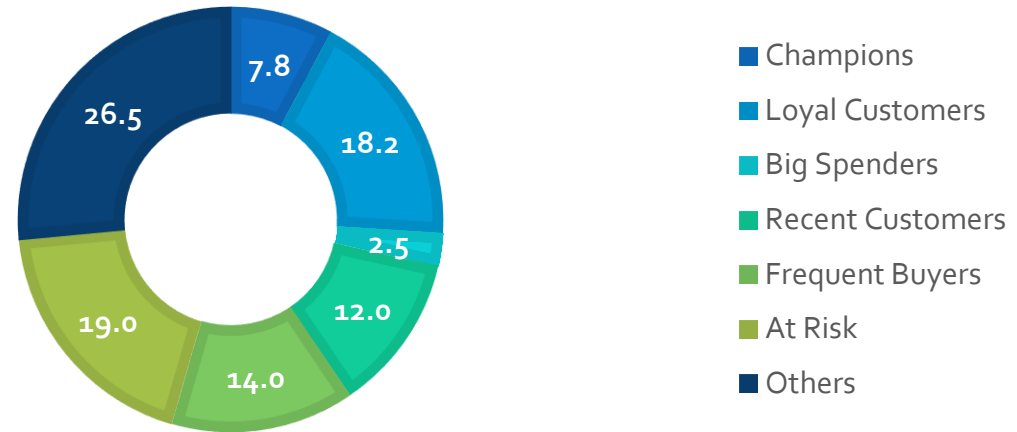
# Distribution of Customer Segments



- The majority fall under “Others” and “At Risk,” while “Champions” and “Big Spenders” are smaller but high-potential groups.

# Percentage Share of Segments

PERCENTAGE DISTRIBUTION OF CUSTOMER SEGMENTS



- Nearly half of customers (45.5%) are in low-engagement categories (“Others” + “At Risk”), suggesting a retention challenge.



# Insights from RFM Segmentation

- **Retention Challenge:** 45% of customers have low engagement (Others, At Risk).
- **Core Strength:** Loyal Customers & Frequent Buyers together form 32% of the base, representing strong repeat business.
- **High-Value Small Groups:** Champions (7.8%) and Big Spenders (2.5%) together account for only 10% but contribute high revenue.
- **Growth Opportunity:** Recent Customers (12%) can be converted into Loyal Customers through targeted nurturing.

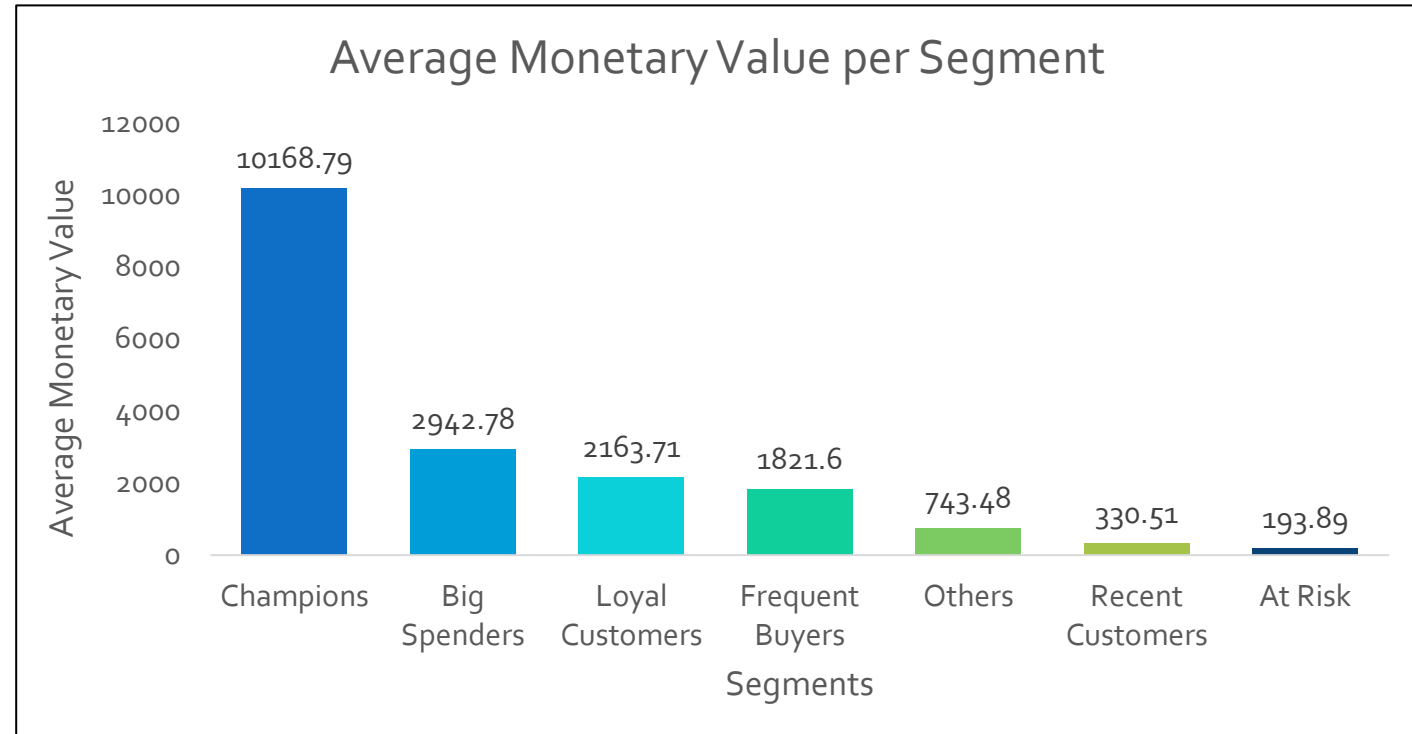
# Recommendations

- **Others & At Risk:** Win-back campaigns, personalized offers, seasonal discounts.
- **Loyal Customers & Frequent Buyers:** Exclusive deals, Loyalty rewards, upselling bundles.
- **Champions:** Early product access, premium offers.
- **Big Spenders:** High-end personalized services.
- **Recent Customers:** Welcome offers, Onboarding and engagement campaigns.

# Conclusion

- Segmentation shows clear behavioral differences in the customer base.
- Large portion is disengaged and requires reactivation strategies.
- High-value customers need personalized retention strategies.
- Predictive modeling can enable real-time segment classification for new customers.

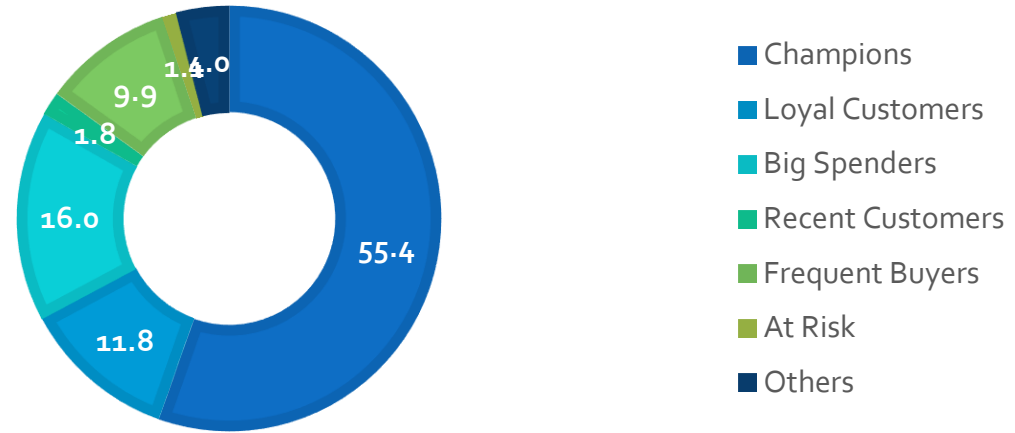
# Average Monetary Value per Segment



- 'Champions' dominate with ₹10,168.79 average spend.
- 'Big Spenders' are second at ₹2,942.78.
- 'Loyal & Frequent ' Buyers have moderate spends.
- 'At Risk' & 'Others' contribute the least.

# Share of Total Monetary Value

PERCENTAGE DISTRIBUTION OF AVERAGE MONETARY VALUE PER SEGMENT



- 'Champions' account for 55.4% of total monetary value.
- 'Big Spenders' and 'Loyal Customers' together make up ~28%.
- Smaller segments offer upselling opportunities.

# Insights from Monetary Analysis

- 'Champions' dominate spending with the highest average monetary value (₹10,168.79).
- 'Big Spenders' contribute significantly (₹2,942.78) despite being a small segment.
- Loyal Customers and Frequent Buyers have moderate average spends, forming a strong revenue base.
- 'At Risk', 'Others', and 'Recent Customers' contribute very little to revenue, highlighting retention challenges.
- Overall, a small percentage of high-value customers drive the majority of revenue.

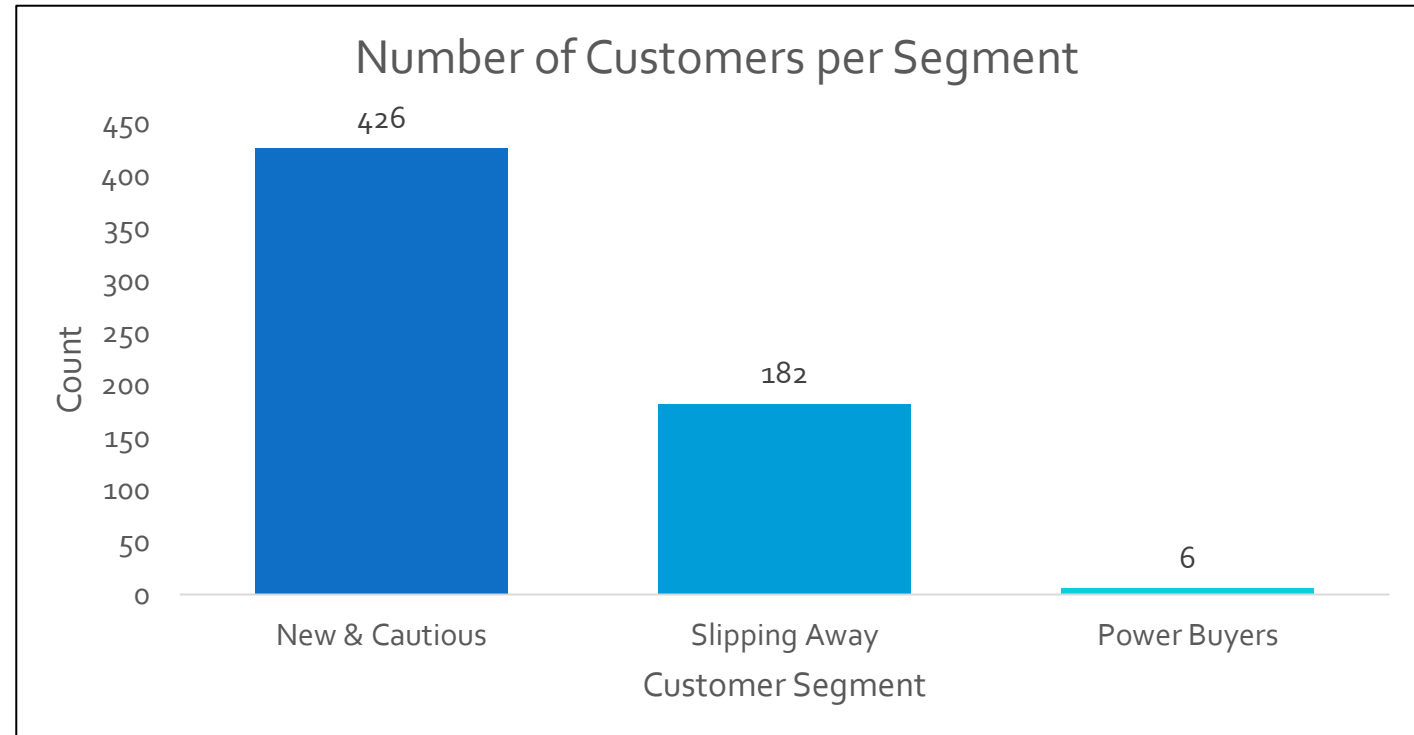
# Clustering Results (K-Means)

| Cluster | Recency | Frequency | Monetary | CustomerCount | Segment        |
|---------|---------|-----------|----------|---------------|----------------|
| 0       | 11      | 1         | 345      | 182           | Slipping Away  |
| 1       | 5       | 1         | 440      | 426           | New & Cautious |
| 2       | 4       | 10        | 9183     | 3             | Power Buyers   |
| 3       | 4       | 4         | 27835    | 1             | Power Buyers   |
| 4       | 7       | 28        | 4074     | 2             | Power Buyers   |

“While RFM gives a good baseline, we further used K-Means clustering to uncover more data-driven patterns.”

- Used Elbow Method & Silhouette Score →  $k = 5$ .
- Cluster 0 → Slipping Away (182 customers).
- Cluster 1 → New & Cautious (426 customers).
- Clusters 2, 3, 4 → Power Buyers (6 customers total).

# Distribution of Clusters



- New & Cautious is the largest group (426).
- Slipping Away group significant at 182 — retention focus needed.
- Power Buyers are small in count (6) but very high in monetary value.



# Actionable Insights on Customer Behavior

- New & Cautious: Recently active but low spend/frequency. High potential for growth.
- Slipping Away: Inactive for longer periods, low spend. At high risk of churn.
- Power Buyers: Small in number but extremely high monetary value and frequency. Critical for revenue

# Marketing Strategies by Segment

- New & Cautious: Welcome campaigns, first-purchase discounts, product bundles, free shipping thresholds.
- Slipping Away: Win-back emails, limited-time offers, loyalty reactivation bonuses, retargeting ads.
- Power Buyers: VIP club, concierge service, premium loyalty tiers, early access to exclusive products.

# Insights & Recommendations

## Insights:

- RFM analysis showed ~45% customers are disengaged (Others, At Risk).
- Loyal Customers & Frequent Buyers (~32%) form the core repeat buyers.
- Champions & Big Spenders (10%) contribute disproportionately high value.
- K-Means clustering (k=5) refined segments into: Slipping Away, New & Cautious, Power Buyers.
- New & Cautious (426) are recent, low-value but high potential customers.
- Slipping Away (182) are inactive and at risk of churn.
- Power Buyers (6) are small but extremely valuable.

## Recommendations (Based on Clustering):

- **New & Cautious:** Welcome campaigns, first-purchase discounts, free shipping thresholds.
- **Slipping Away:** Win-back campaigns, loyalty reactivation bonuses, retargeting ads.
- **Power Buyers:** VIP programs, personalized services, early access to exclusive products.

# Predictive Model Performance

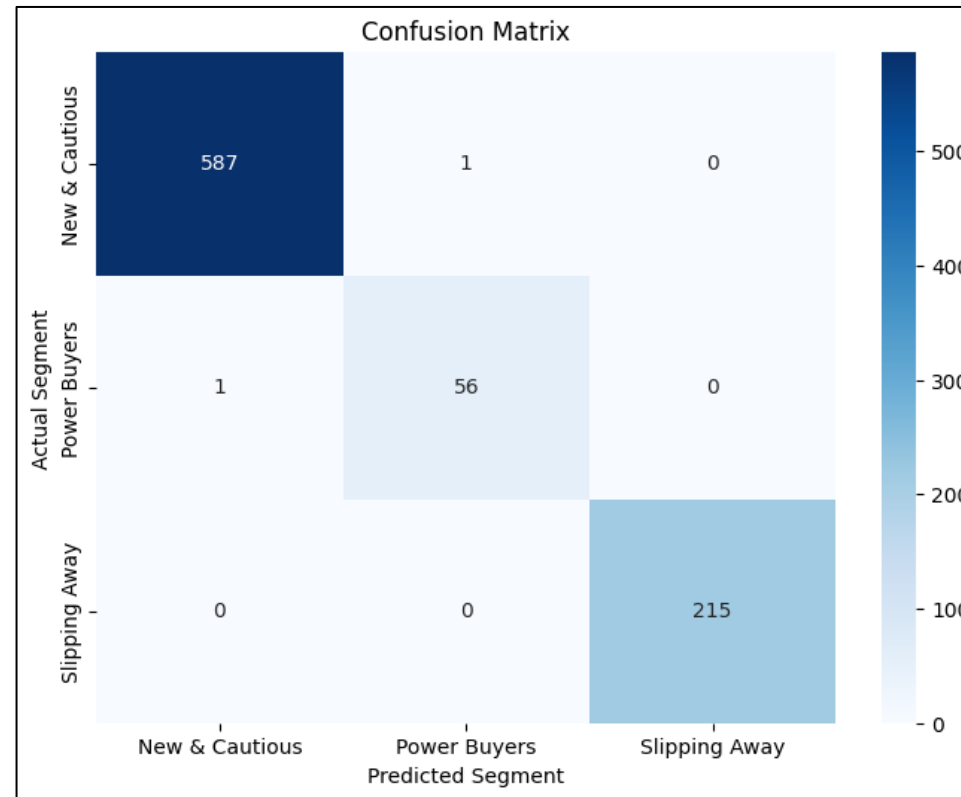
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Classification Report:
              precision    recall  f1-score   support

New & Cautious      1.00      1.00      1.00       588
Power Buyers        0.98      0.98      0.98        57
Slipping Away       1.00      1.00      1.00       215

   accuracy          1.00       860
  macro avg      0.99      0.99      0.99       860
 weighted avg      1.00      1.00      1.00       860
```

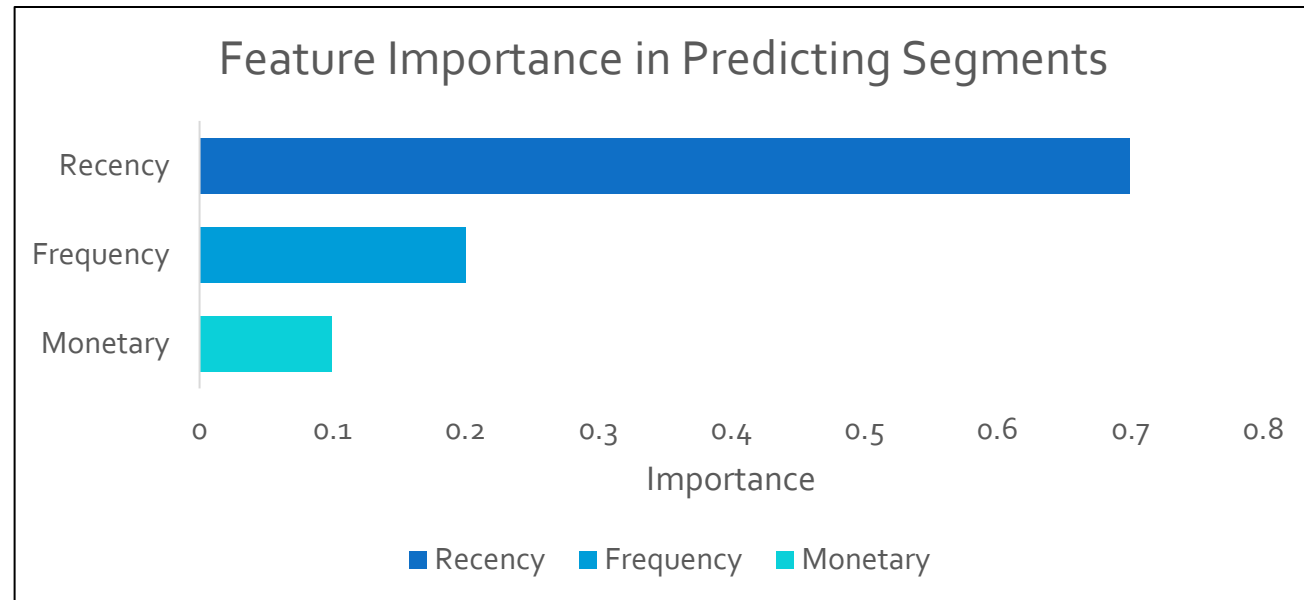
- Model achieved ~100% overall accuracy on test data.
- New & Cautious: Precision, Recall, F1 = 1.00 (perfect classification).
- Slipping Away: Precision, Recall, F1 = 1.00 (perfect classification).
- Power Buyers: Precision, Recall, F1 = 0.98 (minor misclassifications).

# Confusion Matrix



- Confusion Matrix: Only 2 misclassifications out of 860 samples.
- Confusion occurred between New & Cautious vs Power Buyers.
- Almost perfect separation between segments, indicating strong predictive power.

# Feature Importance in Prediction



- Recency (~0.70 importance) is the strongest predictor of customer segment.
- Frequency (~0.20 importance) plays a secondary role, showing repeat purchases matter.
- Monetary (~0.10 importance) has the least influence in prediction.
- Insight: Recent activity is the most critical factor in determining customer behavior.

# Recommendations

- **Customer-Centric Actions (Based on Clustering):**
  - **New & Cautious (426 customers):**
    1. Welcome/onboarding campaigns
    2. First-purchase discounts & free shipping thresholds
    3. Encourage repeat buying with bundles
  - **Slipping Away (182 customers):**
    1. Time-limited offers to re-engage
    2. Loyalty reactivation bonuses
  - **Power Buyers (6 customers):**
    1. VIP programs with exclusive product previews
    2. Personalized concierge service / premium tiers
    3. Early access to limited editions
- **Strategic Focus:**
  - Retain Power Buyers at all costs.
  - Grow New & Cautious into Loyal Customers.
  - Recover Slipping Away before churn becomes permanent.

# Conclusion

- **RFM Analysis:** Gave a clear baseline view of customer engagement → ~45% disengaged, ~10% high-value contributors.
- **Monetary Insights:** Revenue is highly skewed towards a small % of “Champions” & “Big Spenders.”
- **Clustering (K-Means):** Provided refined, data-driven groups → Slipping Away, New & Cautious, Power Buyers.
- **Predictive Model:** Achieved near-perfect accuracy; Recency was the most influential feature.
- **Key Takeaway:**
  - Retention + reactivation efforts can unlock hidden value.
  - Predictive modeling enables real-time classification and personalization.
  - The company should focus resources on retaining Power Buyers while nurturing New & Cautious into repeat buyers.



Thank You